

Possibly Useful Formulae, Chapters 1–11

$$\vec{v} = d\vec{r}/dt; \quad \vec{a} = d\vec{v}/dt$$

$$v_{\text{esc}}(r) = \left(\frac{2GM}{r} \right)^{1/2}$$

$$I_\nu(T) = \frac{2h\nu^3/c^2}{e^{h\nu/kT} - 1}$$

$$\vec{F} = m\vec{a} = d\vec{p}/dt$$

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)$$

$$I_\lambda(T) = \frac{2hc^2/\lambda^5}{e^{hc/\lambda kT} - 1}$$

$$\omega = d\theta/dt = 2\pi/P$$

$$L = I\omega; \quad I = (2/5)MR^2$$

$$L = 4\pi r^2 F$$

$$\frac{1}{P_P} = \frac{1}{P_E} \pm \frac{1}{P_{\text{syn}}}$$

$$\Delta F = \frac{dF}{dr} \Delta r = \frac{2GMm}{r^3} \Delta r$$

$$F = \sigma_{SB} T^4$$

$$\frac{1}{P_{\text{sid}}} = \frac{1}{P_{\text{sol}}} + \frac{1}{P_E}$$

$$r_R = \left(\frac{2M}{m} \right)^{1/3} \Delta r \approx 2.5 \left(\frac{\rho_M}{\rho_m} \right)^{1/3} R$$

$$\theta \approx 1.22 \frac{\lambda}{D}$$

$$\dot{M} = 4\pi r^2 \rho v$$

$$d = a/\tan \pi'' \approx a/\pi''$$

$$r_H = \left(\frac{M_\oplus}{2M_\odot} \right)^{1/3} a_\oplus$$

$$P_B = \frac{B^2}{2\mu_0}$$

$$\vec{F} = \frac{-GMm}{r^2} \hat{r}$$

$$V_{\text{sph}} = \frac{4}{3}\pi R^3; \quad A_{\text{sph}} = 4\pi R^2$$

$$P_{\text{gas}} = \frac{\rho k_B T}{\mu m_p} = nk_B T$$

$$\vec{L} = \vec{r} \times m\vec{v} = mrv_t \hat{\mathbf{k}} = mr^2 \frac{d\theta}{dt} \hat{\mathbf{k}}$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$E = (3/2)NkT$$

$$\frac{dA}{dt} = \frac{L}{2m}$$

$$E = h\nu = hc/\lambda$$

$$W_p = \left(\frac{L_\odot}{4\pi r^2} \right) \pi R^2 (1 - A)$$

$$r = \frac{L^2}{GMm^2(1 + e \cos \theta)}$$

$$\lambda_p \approx \frac{2900 \mu\text{m} K}{T}$$

$$v_{\text{rms}} \lesssim \frac{1}{6} v_{\text{esc}}$$

$$e = \sqrt{(1 - b^2/a^2)}$$

$$\frac{-(Ze)e}{4\pi\epsilon_0 r^2} = \frac{-m_e v^2}{r}$$

$$g = GM/R^2$$

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{Ze^2 m_e} n^2$$

$$\tau \sim \frac{E}{dE/dt}$$

$$r_{\text{ap,pe}} = a(1 \pm e)$$

$$L = m_e v r = n\hbar$$

$$\frac{dP}{dr} = -\frac{GM_r \rho(r)}{r^2} = -\rho g$$

$$P^2 = \frac{4\pi^2 a^3}{GM}$$

$$E_n = -\frac{m_e c^2}{2} \alpha^2 \frac{Z^2}{n^2}; \quad \alpha \approx \frac{1}{137}$$

$$dM_r = 4\pi r^2 \rho(r) dr$$

$$E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

$$H = \frac{kT}{g\mu m_p}$$

$$P_c = \frac{2\pi}{3} \bar{\rho}^2 G R^2$$

$$E = \left(\frac{GMm}{L} \right)^2 \frac{m}{2} (e^2 - 1)$$

$$I(x) = I_0 e^{-n\sigma x}$$

$$\tau = n\sigma x$$

$$U = -\frac{3}{5} \frac{GM^2}{R}$$

Constants

$G = 6.673 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	$u = 1.6605 \times 10^{-27} \text{ kg}$
$\varepsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$	$m_e = 9.109 \times 10^{-31} \text{ kg}$
$\mu_0 = 4\pi \times 10^{-7} \text{ W m}$	$\text{eV} = 1.602 \times 10^{-19} \text{ J}$
$e = 1.602 \times 10^{-19} \text{ C}$	
$c = 2.998 \times 10^8 \text{ m s}^{-1}$	$M_\odot = 1.989 \times 10^{30} \text{ kg}$
$h = 6.626 \times 10^{-34} \text{ J s}$	$L_\odot = 3.84 \times 10^{26} \text{ W}$
$k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$	$R_\odot = 6.955 \times 10^5 \text{ km}$
$\sigma_{SB} = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$	$M_\oplus = 5.974 \times 10^{24} \text{ kg}$
$a = 4\sigma_{SB}/c = 7.56 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$	$R_\oplus = 6378 \text{ km}$
$\sigma_e = 6.652 \times 10^{-29} \text{ m}^2$	$g_\oplus = 9.81 \text{ m s}^{-2}$
$m_p = 1.6726 \times 10^{-27} \text{ kg}$	$1 \text{ pc} = 3.086 \times 10^{13} \text{ km}$
$m_n = 1.674 \times 10^{-27} \text{ kg}$	$1 \text{ AU} = 1.4960 \times 10^{11} \text{ m}$

$$\begin{array}{llll} \text{centi (c)} = 10^{-2} & \text{milli (m)} = 10^{-3} & \text{micro } (\mu) = 10^{-6} & \text{nano (n)} = 10^{-9} \\ \text{kilo (k)} = 10^3 & \text{mega (M)} = 10^6 & \text{giga (G)} = 10^9 & \end{array}$$

$$\text{N (force)} = \text{kg m s}^{-2} \quad \text{J (energy)} = \text{N m} \quad \text{W (power)} = \text{J s}^{-1} \quad \text{pressure} = \text{force/area}$$